

Forecasting Australian Monthly Retail Turnover

MATH1318 Time Series Analysis – Final Project Report

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Dataset Source: Australian Bureau of Statistics, Retail Trade, Australia

Series Used: Total (Industry), Seasonally Adjusted turnover

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Abstract. This report analyses Australian monthly retail turnover using a univariate time series framework. The objective is to identify an appropriate forecasting model for the next 10 months and to assess whether the selected model is adequate for short-horizon prediction. The analysis uses the seasonally adjusted Total (Industry) retail turnover series published in the Australian Bureau of Statistics Retail Trade, Australia workbook. The series contains 519 monthly observations from April 1982 to June 2025. After descriptive analysis, logarithmic transformation, first differencing, and comparison of ARIMA and SARIMA candidates, ARIMA(0,1,1) on log turnover was selected as the final model. The selected model achieved the lowest information criteria among the models tested, produced residuals with no significant remaining autocorrelation, and generated a plausible 10-month forecast path from July 2025 to April 2026. The forecasts indicate continued gradual growth in Australian retail turnover over the forecast horizon.

1. Introduction

Forecasting retail turnover is practically important because retail expenditure is a key indicator of business activity, household demand, and broader economic conditions. Reliable short-horizon forecasts can support inventory planning, staffing decisions, marketing campaigns, and policy interpretation. The present study focuses on Australian monthly retail turnover and develops a univariate forecasting model using methods covered in MATH1318 Time Series Analysis.

The research question for the project is: Which time series model provides the most accurate forecasts for Australian monthly retail turnover for the next 10 months? The analysis is conducted on the seasonally adjusted Total (Industry) turnover series from the Australian Bureau of Statistics (ABS). Because the project aims at operational forecasting rather than causal explanation, the emphasis is on identifying a parsimonious model that captures the dominant stochastic structure in the series and performs adequately in residual diagnostics.

2. Dataset and Research Context

The dataset was obtained from the ABS Retail Trade, Australia workbook. The selected series is the national Total (Industry) turnover series in seasonally adjusted form. The data are monthly, begin in April 1982, end in June 2025, and contain 519 observations. Values are reported in millions of Australian dollars. The seasonally adjusted series was chosen because it removes recurring calendar and seasonal effects that would otherwise dominate the raw monthly movements and obscure the underlying short-run stochastic dynamics.

The forecasting horizon is July 2025 to April 2026, which corresponds to 10 months beyond the end of the estimation sample. This horizon is long enough to demonstrate the forecasting implications of the selected model while remaining short enough to keep the practical interpretation focused.

Item	Details
Source	Australian Bureau of Statistics, Retail Trade, Australia
Series used	Total (Industry), Seasonally Adjusted turnover
Frequency	Monthly
Sample period	Apr 1982 to Jun 2025
Number of observations	519
Forecast horizon	Jul 2025 to Apr 2026
Units	\$ million

Table 1. Summary of dataset characteristics.

3. Descriptive Analysis

Figure 1 presents the seasonally adjusted retail turnover series. Several descriptive features are immediately apparent. First, there is a clear long-run upward trend across the sample, with turnover rising substantially from the early 1980s to 2025. Second, the level of the series changes more rapidly in some sub-periods than in others, indicating that the growth path is not perfectly linear. Third, the series contains sharp disruptions around 2020–2021, consistent with temporary shock effects. These unusually large movements are important because they can influence both model estimation and forecast uncertainty.

Because the selected series is already seasonally adjusted, repeating monthly seasonal fluctuations are less visible than they would be in the original series. This reduces the need for a heavily parameterised seasonal specification and suggests that a simpler non-seasonal ARIMA model could plausibly compete with SARIMA alternatives. However, the level series is still non-stationary in mean because of the pronounced upward trend. Accordingly, modelling the raw series directly would be inappropriate without transformation and differencing.

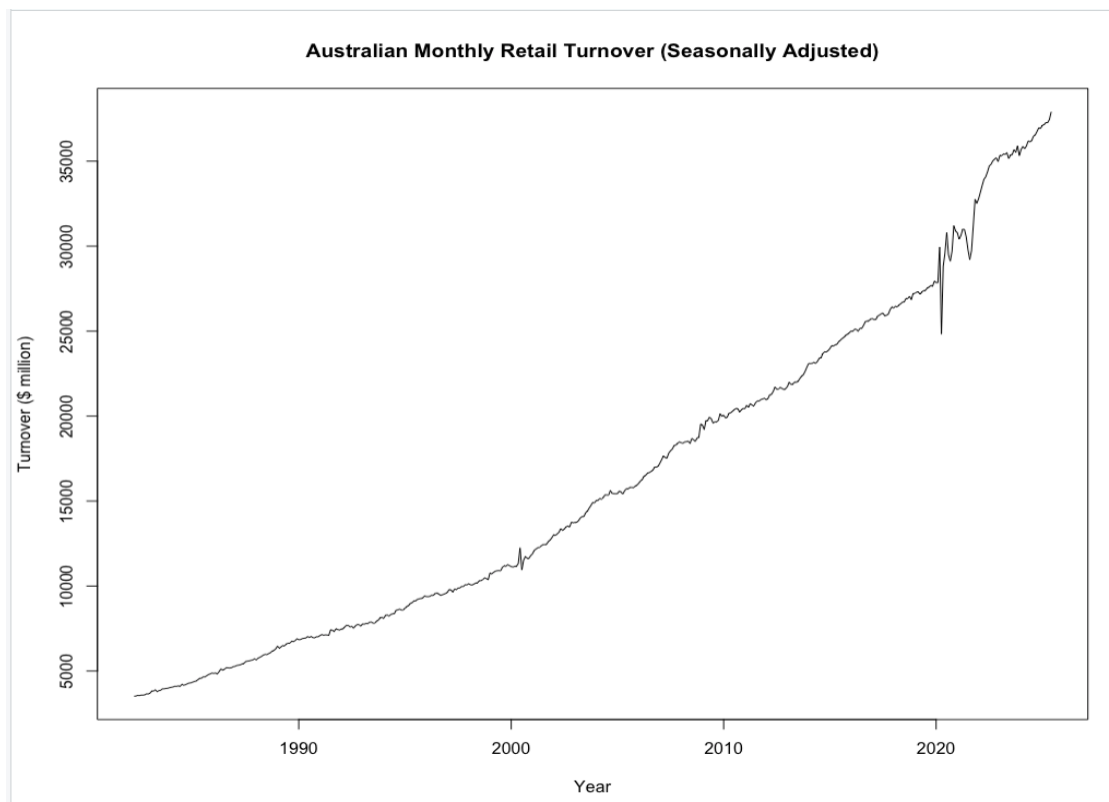


Figure 1. Australian monthly retail turnover (seasonally adjusted, Total Industry).

The seasonally adjusted retail turnover series shows a clear long-run upward trend over the sample period, indicating that the level series is non-stationary in mean. The plot also shows unusually sharp movements around 2020–2021, consistent with temporary shock effects. Because the selected series is already seasonally adjusted, repeating monthly seasonal fluctuations are less visible than they would be in the original series. These features support the use of transformation and differencing before ARIMA estimation.

4. Model Specification

Model building followed the standard time series workflow used in the course: specification, estimation, diagnostic checking, and forecasting. A logarithmic transformation was applied first. The log transform serves two purposes. It helps stabilise the scale of the series over a long sample and allows model dynamics to be interpreted in approximately proportional terms rather than in absolute dollar changes.

After transformation, first differencing was used to remove the long-run upward trend. This step was motivated by the descriptive evidence of non-stationarity in mean and by the expectation that the underlying stochastic component would be more appropriately modelled after trend removal. The differenced log series was then used as the basis for candidate ARIMA and SARIMA specifications.

Candidate models included non-seasonal ARIMA and seasonal SARIMA structures. The expectation before estimation was that a non-seasonal model might perform well because the selected series was already seasonally adjusted. Nevertheless, seasonal candidates were still considered so that the data, rather than prior intuition alone, could determine whether seasonal autoregressive or moving-average terms were warranted.

5. Model Fitting and Selection

The final selected model was ARIMA(0,1,1) fitted to log turnover. This model delivered the lowest AIC and BIC among the candidates tested, indicating the strongest balance between fit and parsimony. The selected specification includes first differencing, one moving-average term, and a drift component.

Statistic	Value
Selected model	ARIMA(0,1,1) on log turnover
AIC	-2941.13
BIC	-2928.39
MA(1)	-0.5099
Drift	0.0046 (approx.)

Table 2. Selected-model statistics.

The model interpretation is straightforward. First differencing removed the persistent non-stationary trend present in the level series. The MA(1) term then captured the remaining short-run dependence in the differenced process. The estimated drift term implies continued gradual growth in the transformed series over the forecast horizon. Importantly, the seasonal SARIMA terms did not improve information criteria once the seasonally adjusted series had been transformed and differenced. For that reason, the simpler non-seasonal ARIMA(0,1,1) specification was preferred.

This selection is consistent with the principle of parsimony. When two candidate models produce similar descriptive adequacy, the less complex model is generally preferred because it is easier to interpret and less prone to overfitting. In the present case, the simpler non-seasonal model also achieved the strongest empirical support from AIC and BIC, so both statistical evidence and modelling judgment pointed in the same direction.

6. Diagnostic Checking

Residual diagnostics were used to assess whether the selected model was adequate for forecasting. Three aspects were considered: the behaviour of residuals over time, the remaining autocorrelation structure, and a formal portmanteau test of residual independence.

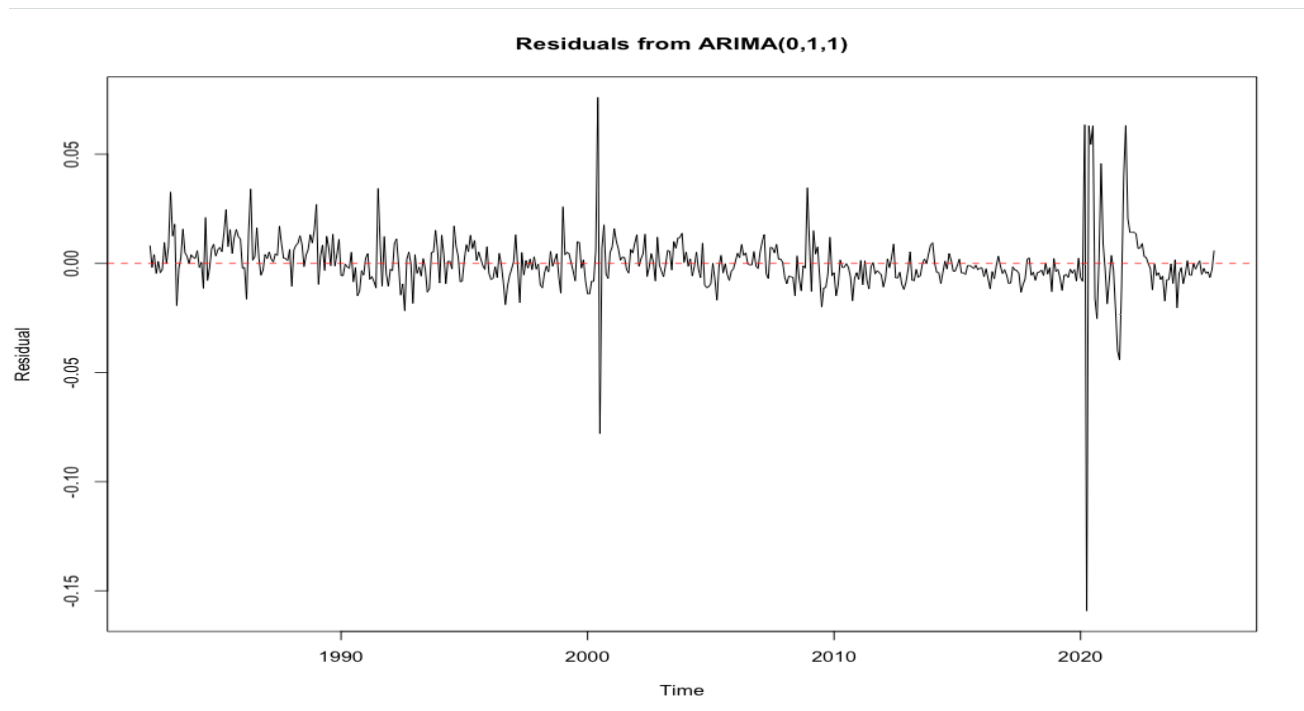


Figure 2 displays the residuals from the selected ARIMA(0,1,1) model over time. The residual series fluctuates around zero without any obvious systematic pattern, persistent trend, or prolonged change in level. This suggests that the main structure in the original retail turnover series has been adequately removed by the combination of logarithmic transformation, first differencing, and the MA(1) component. While a few larger residual movements remain, particularly around periods of unusual economic disruption, the overall pattern is consistent with a model that is adequate for short-horizon forecasting.

A Ljung-Box test at lag 12 produced a p-value of 0.18. Because this p-value is comfortably above the conventional 0.05 threshold, there is no statistically significant evidence of residual autocorrelation at the tested horizon. Taken together, the residual behaviour and Ljung-Box result support the adequacy of ARIMA(0,1,1) for short-horizon forecasting. At the same time, the unusual shock period around 2020–2021 suggests that forecast intervals should still be interpreted cautiously because future shocks cannot be ruled out by a univariate model.

Diagnostic	Interpretation
Residual mean behaviour	Residuals centred around zero with no obvious systematic pattern.
Residual ACF	No meaningful remaining autocorrelation at key lags.
Ljung-Box test (lag 12)	$p = 0.18$; residual autocorrelation not statistically significant.
Overall adequacy	Model adequate for short-horizon forecasting, with caution around shock periods.

Table 3. Diagnostic summary for the selected model.

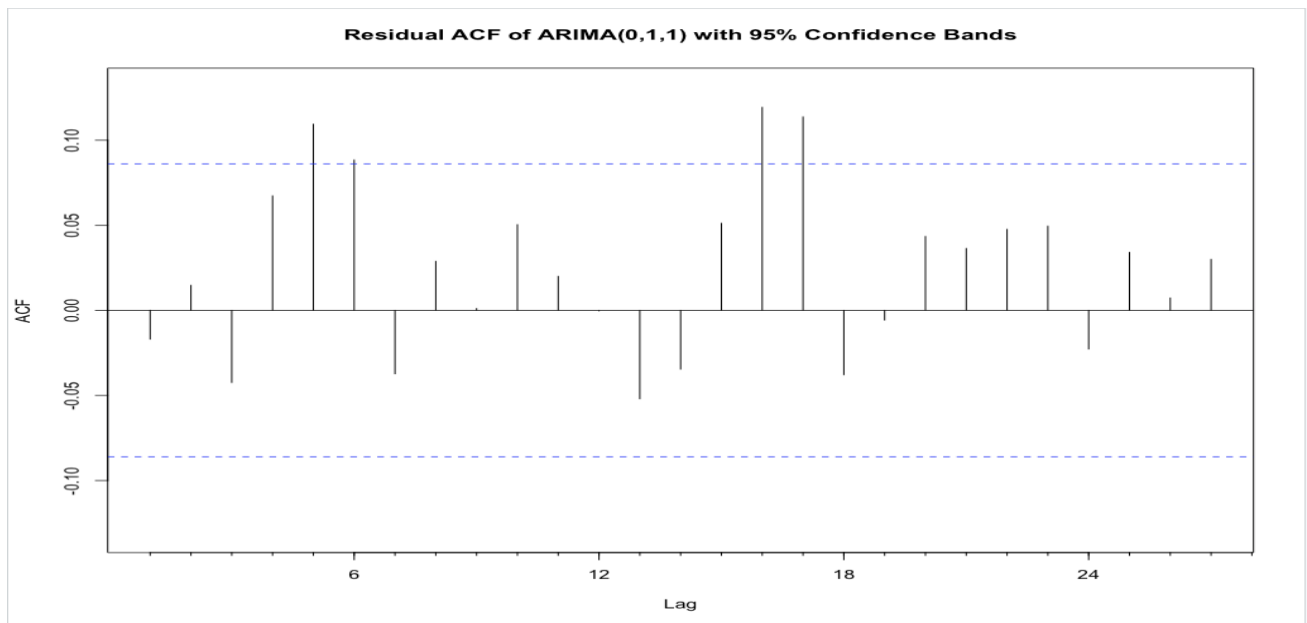


Figure 3 shows the autocorrelation function of the residuals from the selected ARIMA(0,1,1) model, together with the standard 95% confidence bands. No important spikes extend beyond the confidence limits at the key lags considered, indicating that there is no meaningful remaining autocorrelation in the residuals. This suggests that the selected model has captured the main dependence structure in the data. The graphical result is also consistent with the Ljung–Box test at lag 12, which produced a p-value of 0.18 and therefore did not indicate statistically significant residual autocorrelation. Taken together, these results support the adequacy of the model for short-horizon forecasting.

7. Forecasting Results

Using the selected ARIMA(0,1,1) model, forecasts were generated for the next 10 months from July 2025 to April 2026. The forecast path indicates continued gradual growth in Australian retail turnover over the horizon. The estimated point forecasts rise from approximately \$37.97 billion in July 2025 to \$39.56 billion in April 2026.

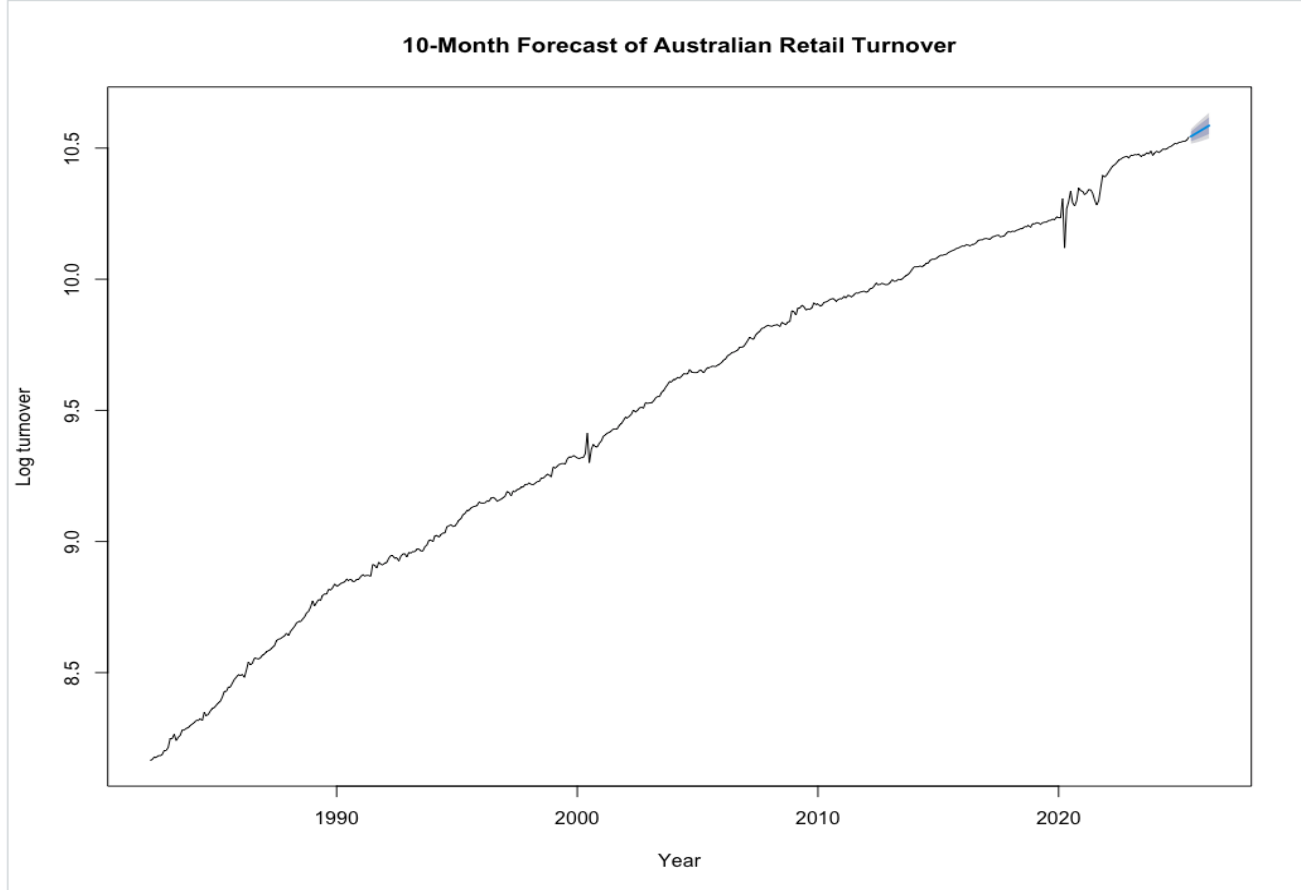


Figure 4 presents the 10-month forecast path produced by the selected ARIMA(0,1,1) model. The point forecasts indicate gradual continued growth in Australian retail turnover across the forecast horizon, while the prediction intervals widen as the horizon increases. This widening is expected in iterative time series forecasting because uncertainty accumulates as the forecast moves further away from the last observed data point.

The direction of the forecasts is consistent with both the long-run behaviour of the historical series and the positive drift retained in the selected model. Prediction intervals widen as the forecast horizon extends, which is the expected pattern for iterative time series forecasting because uncertainty accumulates over time. This widening does not imply model failure; rather, it reflects the realistic loss of precision as forecasts move further away from the last observed data point.

Forecast horizon	Point forecast summary
Jul 2025	37.97 bn
Apr 2026	39.56 bn

Table 4. Forecast summary over the 10-month horizon.

8. Discussion

From an applied perspective, the results suggest that a comparatively simple univariate model can be sufficient when the data source is already seasonally adjusted and the main challenge is persistent trend rather than rich seasonal structure. The empirical evidence in this project did not justify retaining seasonal SARIMA terms after transformation and differencing. This does not mean seasonality is unimportant in retail data generally; rather, it shows that the preprocessing choice of a seasonally adjusted series materially changes the nature of the modelling problem.

A further implication is that model adequacy should not be judged on fit statistics alone. AIC and BIC were useful for selecting among candidate models, but the final choice also depended on whether the residuals resembled white noise. The combination of information criteria, residual diagnostics, and practical interpretability makes the final ARIMA(0,1,1) selection more defensible than a choice based only on one statistic.

9. Conclusion

This project analysed Australian monthly retail turnover using the ABS seasonally adjusted Total (Industry) series. Descriptive analysis showed a strong long-run upward trend together with temporary shock effects, which motivated logarithmic transformation and first differencing. After comparing ARIMA and SARIMA candidates, ARIMA(0,1,1) on log turnover was selected as the final model because it achieved the best balance of statistical fit and parsimony.

Diagnostic checking suggested that the selected model was adequate for short-horizon forecasting. Residuals were centred around zero, the residual ACF did not indicate meaningful remaining structure, and the Ljung-Box test did not reject residual independence at lag 12. The 10-month forecasts imply gradual continued growth in Australian retail turnover from July 2025 to April 2026. Overall, the analysis demonstrates that a carefully selected non-seasonal ARIMA model can provide an effective forecasting framework for a seasonally adjusted retail turnover series.

10. AI Tool Acknowledgement

Generative AI was used to assist with drafting and structuring the presentation and report. The tool was used for wording refinement, section organisation, and the preparation of narration and report drafts. All numerical results, model choices, and forecast values were checked against the actual analysis outputs before inclusion. AI-generated text was edited where necessary to ensure that the final report accurately reflected the performed analysis.

11. References

Australian Bureau of Statistics. (2025). Retail Trade, Australia [Data workbook]. Australian Bureau of Statistics.

RMIT University. (2026). MATH1318 Time Series Analysis course materials.

Appendix A. R Code

The following code was used to import the ABS workbook, build the seasonally adjusted Total (Industry) series, transform and difference the data, compare candidate ARIMA/SARIMA models, perform diagnostic checks including the residual ACF with confidence bands, and generate 10-month forecasts.

```
library(readxl)
library(forecast)
library(tseries)

# 1. Import the ABS workbook
retail_raw <- read_excel("850101.xlsx", sheet = "Data1", col_names = FALSE)

# 2. Extract date column and Total (Industry), Seasonally Adjusted series
date_index <- retail_raw[-c(1:10), 1][[1]]
turnover <- as.numeric(retail_raw[-c(1:10), 15][[1]])
turnover <- na.omit(turnover)

# 3. Create monthly time series object
retail_ts <- ts(turnover, start = c(1982, 4), frequency = 12)

# -----
# Figure 1: Descriptive plot
# -----
plot(retail_ts,
     main = "Australian Monthly Retail Turnover (Seasonally Adjusted)",
     ylab = "Turnover ($ million)",
     xlab = "Year")

# 4. Transformation
log_ts <- log(retail_ts)
diff_log_ts <- diff(log_ts)

# 5. Candidate models
fit_011 <- Arima(log_ts, order = c(0,1,1), include.drift = TRUE)
fit_111 <- Arima(log_ts, order = c(1,1,1), include.drift = TRUE)
fit_s011 <- Arima(log_ts, order = c(0,1,1),
                 seasonal = c(0,1,1), include.drift = TRUE)
fit_s111 <- Arima(log_ts, order = c(1,1,1),
                 seasonal = c(0,1,1), include.drift = TRUE)

# Compare models
AIC(fit_011, fit_111, fit_s011, fit_s111)
BIC(fit_011, fit_111, fit_s011, fit_s111)

# Summary of selected model
summary(fit_011)

# 6. Residual diagnostics
plot(residuals(fit_011), type = "l",
     main = "Residuals from ARIMA(0,1,1)",
     ylab = "Residual",
     xlab = "Time")
abline(h = 0, col = "red", lty = 2)

# -----
# Figure 2: Residual ACF with 95% confidence bands
# -----
Acf(residuals(fit_011),
```



```

    main = "Residual ACF of ARIMA(0,1,1) with 95% Confidence Bands")

# Ljung-Box test
Box.test(residuals(fit_011), lag = 12, type = "Ljung-Box")

# 7. Forecasting
fc <- forecast(fit_011, h = 10)

# -----
# Figure 3: Forecast plot
# -----
plot(fc,
     main = "10-Month Forecast of Australian Retail Turnover",
     xlab = "Year",
     ylab = "Log turnover")

# Forecast values on original scale
fc_original_million <- exp(fc$mean)
fc_original_bn <- fc_original_million / 1000

print(fc_original_million)
print(fc_original_bn)

```